Data-dependent inference & instrument hacking

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01 Data-dependent inference

Letting the data shape our analysis choices is hacky behavior in more ways than we realize

02 Selective inference

Everybody knows post-selection inference is hacky behavior

03 Instrument hacking

Those rich enough to have multiple instruments are also probably engaging in hacky behavior

01 Data-dependent inference

Data-dependent inference is an expanding umbrella describing the "overreach" of the credibility revolution

The number of problems continues to grow faster than we can come up with solutions

PROBLEMS

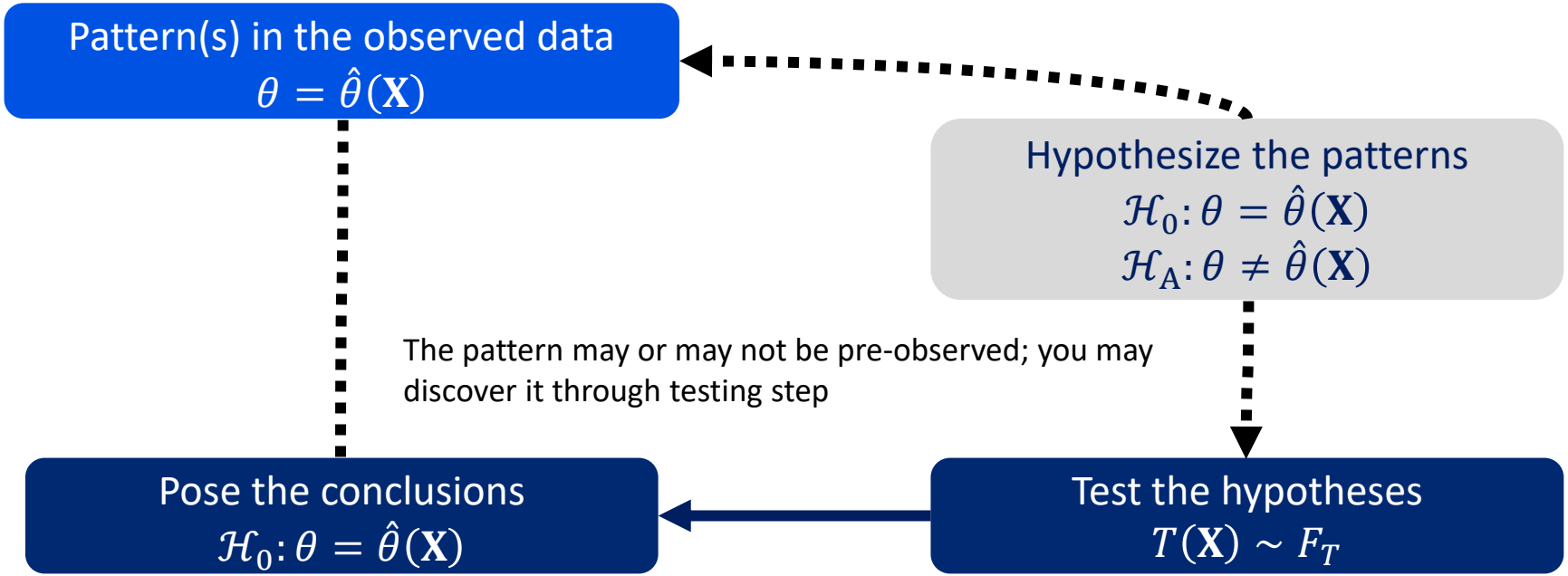
- Publication bias
- Multiple testing problem
- Post hoc theorizing/post hoc analysis
- HARKing
- p-hacking
- Selective inference
- Instrument hacking

SOLUTIONS

- Multiple testing corrections
- Scheffe's method
- Cross validation
- Confirmation sampling
- Knock-off filters
- "Assumption-lean"/robust inference

Many more solutions to (hopefully) be had!

Data-dependent inference is problematic because data drives the hypothesis and analysis (double dipping)



02 Selective inference

Selective inference is an old idea at the foundation of elementary machine learning

Fundamental problem (plain English)

Conclusions on data drawn from a model favored by that data are only informative on that data

Fundamental problem (formal)

The null distribution of some test statistic $T(\boldsymbol{X})$ pre-selection is different from the null distribution of the test statistic $T(\boldsymbol{X})$ post-selection:

For some selection event $E(\boldsymbol{X})$, $T(\boldsymbol{X}) \sim_{H_0} F$, $T(\boldsymbol{X})|E(\boldsymbol{X}) \sim_{H_0} G$ but usually $F \neq_{\mathcal{D}} G$

Example: selection on the “winner” demonstrates the degree to which selection can distort inference

$X_i \sim \mathcal{N}(0,1)$ IID for $i \in [n]$.

Define

$$X_j = \max_{i \in [n]} X_i = 2$$

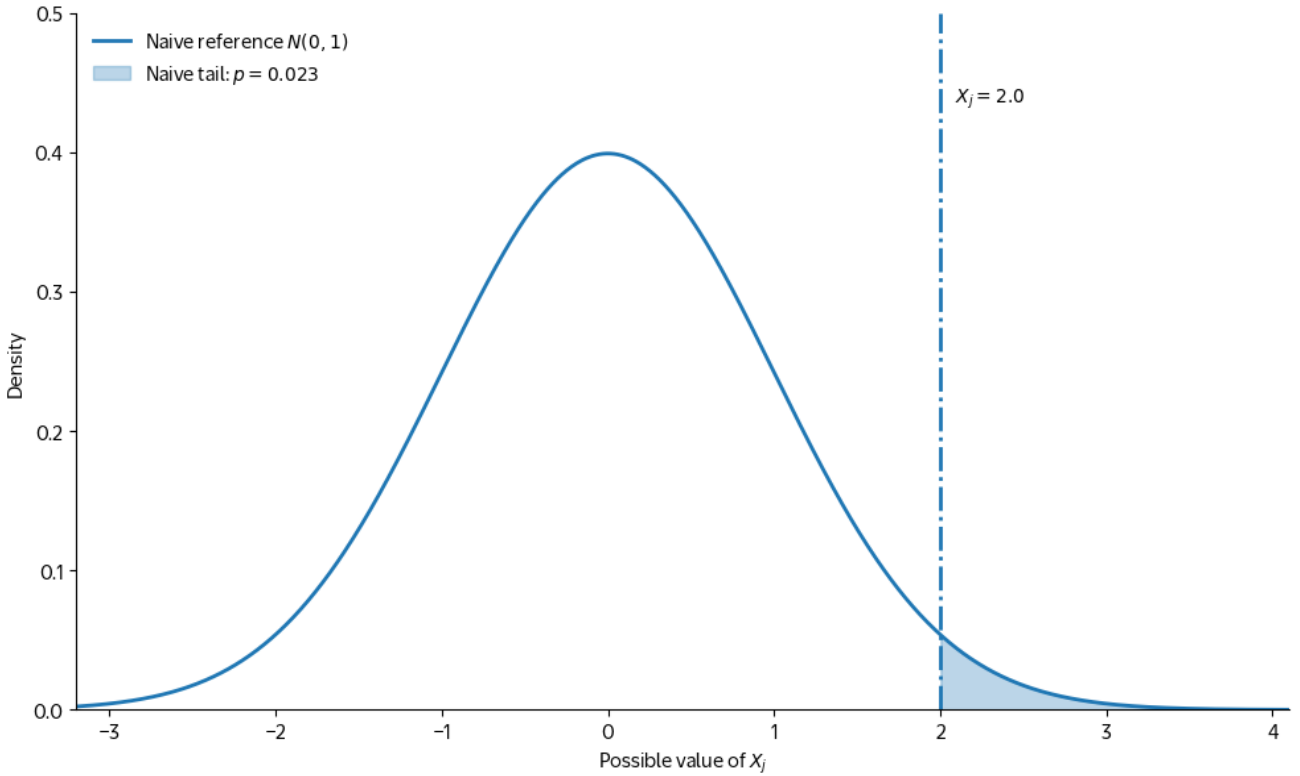
$$j = \text{argmax}_{i \in [n]} X_i$$

Inference target

$$\text{H}_0 : \mathbb{E}[X_j] = 0$$

We might use X_j as the test-statistic & suppose $\mathbb{V}(X_j) = 1$ known…

$$p \approx 0.023$$



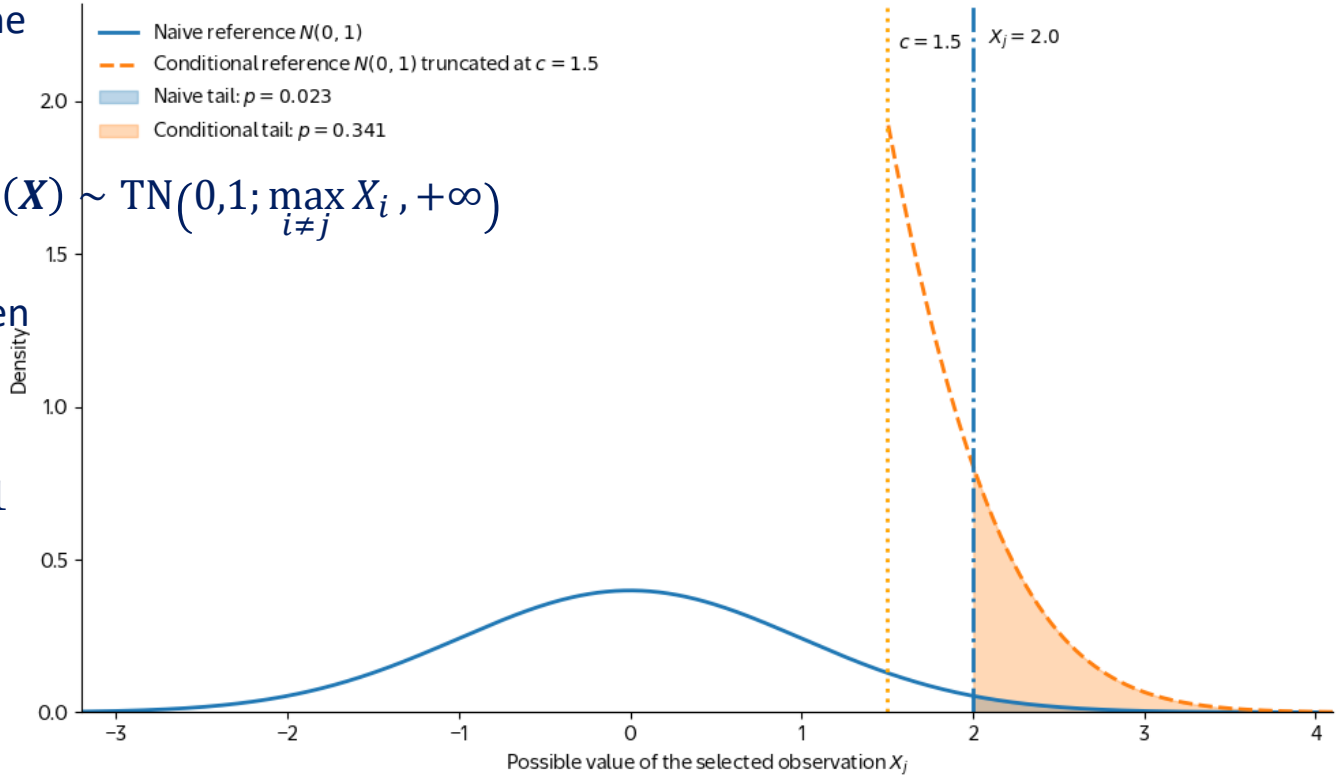
Example: selection on the “winner” demonstrates the degree to which selection can distort inference

We need to condition on the selection event!

$$X_j | j = \text{argmax}_{i \in [n]} X_i, M_{e_j}(\boldsymbol{X}) \sim \text{TN}\left(0,1; \max_{i \neq j} X_i, +\infty\right)$$

Suppose $\max_{i \neq j} X_i = 1.5$. Then our new p-value is then:

$$p = \frac{1 - \Phi(2)}{1 - \Phi(1.5)} \approx 0.341$$



Conditioning on selection is just one way to conduct valid inference under selection

There are several other valid inferential frameworks for dealing with selection, depending on the type of selection (e.g., OLS, LASSO, inference on the winner, etc.):

- Sample splitting
- Cross-validation
- Knockoff covariates
- And more!

03 Instrument hacking

Some notation before a quick refresher of inference for instrumental variables…

Assume IID population $i \in [n]$ in fixed $\boldsymbol{X}_i = (\boldsymbol{D}_i, \boldsymbol{W}_i)$ and $\boldsymbol{F}_i = (\boldsymbol{Z}_i, \boldsymbol{W}_i)$, random Y_i, and linear IV model framework

- $\boldsymbol{F}_i$ matrix of excluded variables with $\dim(\boldsymbol{F}_i) \geqslant 1$, $\boldsymbol{X}_i$ matrix of included variables
- $\boldsymbol{D}_i$ matrix of endogenous variables
- $\boldsymbol{W}_i$ covariate matrix

Matrix of weights $\boldsymbol{C}$ with $\dim(\boldsymbol{C}) = \dim(\boldsymbol{F}_i)$ to select instruments when $\dim(\boldsymbol{F}_i) > \dim(\boldsymbol{X}_i)$

- This is useful for our refresher—we will assume $\dim(\boldsymbol{D}_i) = 1$ so $\boldsymbol{C} = \mathbf{1}_{\dim(\boldsymbol{W}_i) \times 1}$ for simplicity moving forward

(Nick please excuse some abuse of notation relative to the paper for convenience)



01

Standard t inference

For consistently estimated weights matrix $\boldsymbol{C} \rightarrow_p \boldsymbol{c}$, a sufficiently relevant instrument in a linear IV model has approximately:

$$\boldsymbol{\beta}_{IV}(\boldsymbol{C}) \approx_d \mathcal{N}\left(\boldsymbol{\beta}, \frac{1}{n}\boldsymbol{\Sigma}_n(\boldsymbol{c})\right)$$

02

Anderson-Rubin inference

Even with weak instrument(s) we can test and invert $H_0: \boldsymbol{\beta}^{\mathrm{RF}} = \boldsymbol{\beta}_0$. Can do this parametrically or via simulation (simulation is surprisingly robust re: type I error & power)!

Given K many possible instruments, how is the researcher to choose one to use?

A researcher unknowingly has available to them K possible instruments (relevant and plausibly exogenous and excluded) to deal with scalar endogenous variable D_i:

$$\mathcal{Z}_i = \{Z_{1i}, \dots, Z_{Ki}\}$$

- $\mathcal{Z}$ can be a treasure chest or a trash can
  - Sometimes we may actually have/come up with $\mathrm{K} > 1$ viable instruments; sometimes we may have $\mathrm{K} > 1$ instruments out of necessity/weak economic narrative

Q: How should a researcher "select" their instrument(s)?

- Reminder: we really only need 1 instrument to be exactly/just ID'd!

Some instruments (and a lot of the really good ones) are given to man by God



The blessed economist might:

- Use an economic narrative to choose/create an instrument
- Have access data or institutions with excluded exogenous variables

If the instrument works, its simplicity is usually very elegant!

(Don't judge my taste I thought this instrument was clever af when I first learned about it)

Other instruments are just the least unpalatable dishes at the nasty buffet



The unfortunate (or very fortunate) economist might:

- Be able to identify and collect data on several instruments from a motivating structural economic model
- Try to find a rough diamond in a bucket full of dirty glass (a.k.a. something that works)

Given K many possible instruments, how is the researcher to choose one to use?

A researcher unknowingly has available to them K possible instruments (relevant and plausibly exogenous and excluded) to deal with scalar endogenous variable D_i:

$$\mathcal{Z}_i = \{Z_{1i}, \dots, Z_{Ki}\}$$

God did not appear to the researcher in a flaming bush. So the researcher lets the first stage regression/test statistic (FSFS)

$$D_i = \pi_k Z_{ki} + \boldsymbol{W}_i'\boldsymbol{\gamma} + V_{ki}$$

Or second stage regression/test statistic (SSTS)

$$Y_i = \beta \hat{D}_{ki} + \boldsymbol{W}_i'\boldsymbol{\gamma} + U_i$$

Or both chose the most "relevant" instrument Z_{k^*i}

Isn't this reminiscent of some suspicious hacky behavior from earlier?

(yes, it is)

Instrument hacking is when a researcher selects instruments on first-stage, second-stage, or joint statistics

First-stage instrument hacking

- From $\mathrm{D}_i = \pi^k Z_i^k + \boldsymbol{W}_i{}'\boldsymbol{\gamma} + V_i^k$, choose Z^k based on $k^* = \underset{k\in[K]}{\operatorname{argmax}} F_k$ or $k^* = \underset{k\in[K]}{\operatorname{argmax}} \hat{\pi}^k$

Second-stage instrument hacking

- From $Y_i = \beta \hat{D}_i^{\,\mathrm{k}} + \boldsymbol{W}_i{}'\boldsymbol{\gamma} + U_i$, choose Z^k based on $k^* = \underset{k\in[K]}{\operatorname{argmax}} |\mathrm{t}_k|$

Joint instrument hacking $k^* = \underset{k\in[K]}{\operatorname{argmax}} s_k$

- Smooth joint preferences (theoretically)

$$s_k = \lambda F_k + (1-\lambda)|t_k| \text{ for } \lambda \in (0,1)$$

- Lexicographic-threshold preferences (in practice)

$$s_k = (\mathbf{1}\{F_k \geqslant c\}, |t_k|)$$

Instrument selection leads to both estimation and inference problems, even under traditionally robust frameworks

Define the residualized instrument $\tilde{Z}_{ki} = \boldsymbol{M}_{\boldsymbol{W}_i} Z_{ki}$ and $V_{ki} = \rho U_i$ for some endogeneity parameter $\rho \in (-1, 1)$ (i.e., $\rho = 0$ is exogeneity). For simplicity assume $Z_{ki} \perp Z_{\ell i} \mid \boldsymbol{W}_i$.

1. Instrument hacking produces 2SLS estimates biased toward OLS (Keane & Neal 2024)

This first stage coefficient on instrument Z_{ki}:

"true" instrument strength

covariance between instrument k & structural error term U_i

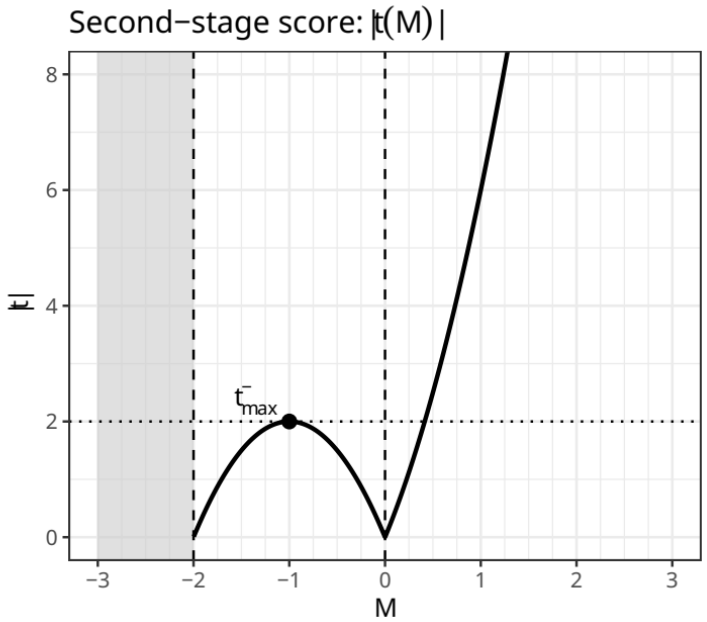
$$\hat{\pi}_{\mathrm{k}} = \frac{\pi_{\mathrm{k}} \hat{\sigma}_k^2 + \rho \hat{\mathbb{C}}\left(\tilde{Z}_{ki}, U_i\right)}{\hat{\sigma}_k^2}$$

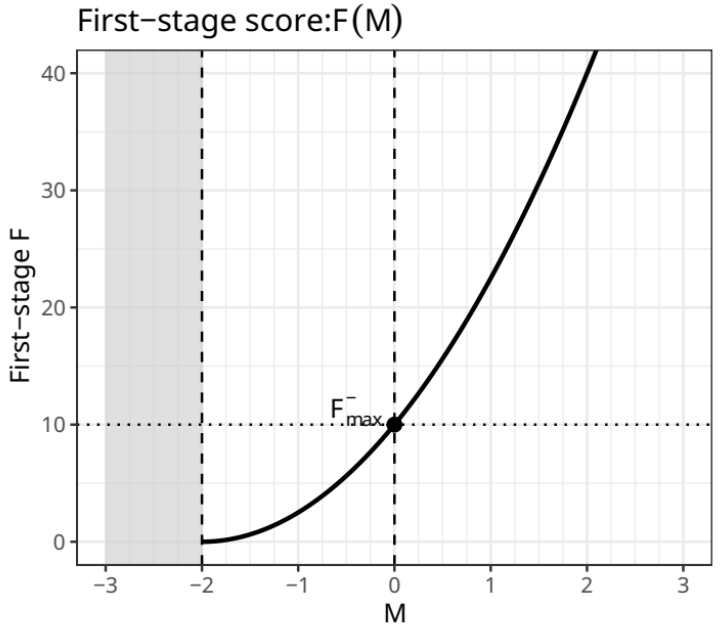
We have some problems when…

- $\hat{\mathbb{C}}(Z_{ki}, U_i) > 0$, i.e., instrument k appears endogenous in finite-samples (which is unobservable and possible) thus appearing a better instrument AND $\widehat{\sigma_k^2}$ is "favorable"

Instrument hacking produces IV estimates biased toward OLS (cont., Keane & Neal 2024)

$$t_k \propto \frac{|\rho\hat{\mathbb{C}}(\tilde{Z}_{ki}, U_i)|(\pi_k\hat{\sigma}_k^2 + \rho\hat{\mathbb{C}}(\tilde{Z}_{ki}, U_i))}{\pi_k\hat{\sigma}_k^2} = M \qquad F_k \propto \left(\pi_k\hat{\sigma}_k^2 + \rho\hat{\mathbb{C}}(\tilde{Z}_{ki}, U_i)\right)^2$$





Instrument hacking produces IV estimates biased toward OLS (cont., Keane & Neal 2024)

Recall the purpose of IV is to remove endogeneity bias $\mathbb{C}(D_i, U_i)$ by using some exogenous variation from Z_i, which ideally has $\mathbb{C}(Z_i, U_i) = 0$

- But instrument hacking tends to favor instruments k with larger, positive $\mathbb{C}(Z_{ki}, U_i)$ (previous slide)
- Using the "best" instrument after some hacking usually means we're selecting one that doesn't cleave out all the endogeneity bias we want



Instrument selection leads to both estimation and inference problems, even under traditionally robust frameworks

2. As a form of selection, instrument hacking distorts inference

- From Gauss-Markov we infer $\hat{\sigma}^2_{2SLS} \geqslant \hat{\sigma}^2_{OLS}$, which we then use to estimate $se(\hat{\beta}^{2SLS})$ and $se(\hat{\beta}^{OLS})$, respectively. t-inference is distorted by bias of β and $\hat{\sigma}^2$ estimates toward OLS. This "power asymmetry" leads to "imbalanced rejections" (more rejections occur to the right of the null than to the left).
  - Use AR instead because it is surprisingly robust! AR reduces median bias but still suffers from size inflation and only mitigates power asymmetry for second-stage instrument hacking

TABLE 3 – 2SLS with AR-test Under Instrument Selection

Type of Hacking:	Second-Stage			First-Stage			Lexicographic		
Instruments Available	Size	Pos. Rej.	Median Bias	Size	Pos. Rej.	Median Bias	Size	Pos. Rej.	Median Bias
1	0.051	52%	0.002	0.051	52%	0.002	0.051	52%	0.002
2	0.099	52%	0.021	0.051	65%	0.020	0.079	60%	0.035
3	0.144	51%	0.047	0.052	70%	0.025	0.112	61%	0.070
5	0.226	51%	0.091	0.054	75%	0.030	0.179	60%	0.120
8	0.337	52%	0.123	0.058	79%	0.033	0.270	61%	0.160
10	0.401	52%	0.137	0.059	81%	0.035	0.326	61%	0.178

Instrument selection leads to both estimation and inference problems, even under traditionally robust frameworks

FIGURE 6. Instrument-Hacking with Instrument Strengths Drawn from Data: Joint Distribution of $\hat{\beta}$ and Standard Errors

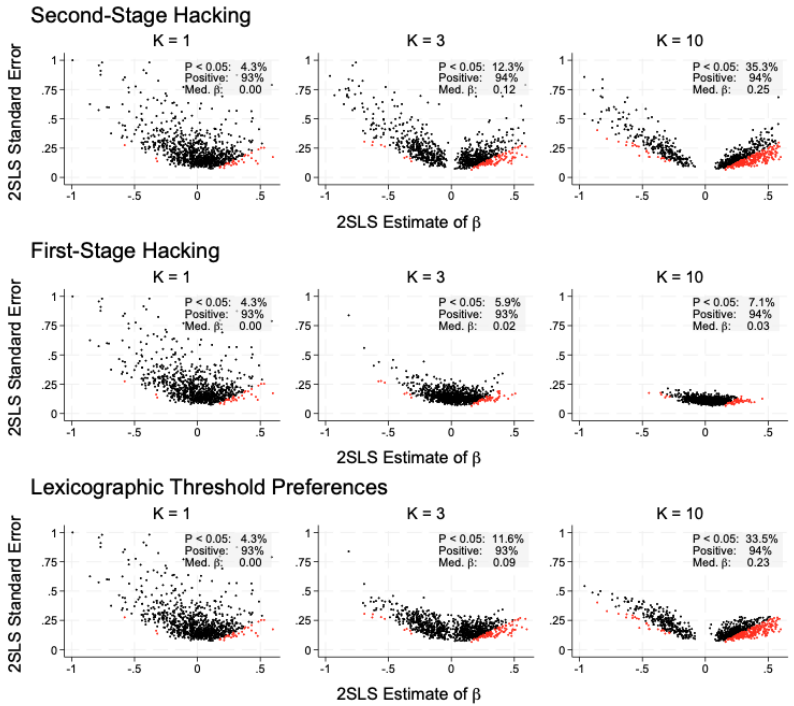
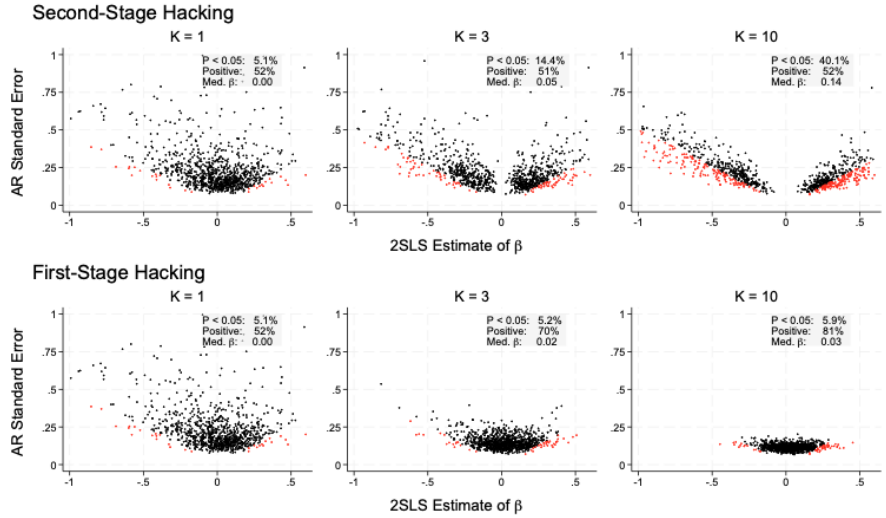


FIGURE 7. Instrument-Hacking with the AR Test and Instrument Strengths Drawn from Data: Joint Distribution of $\hat{\beta}$ and AR Standard Errors



01

Angrist-Kolesar (2024) sign screening test

FS instrument hacking can be fine ~~if we call it "pre-screening"~~ in exactly ID'd cases if you only screen on the sign of the first stage agreeing with a pre-specified economic narrative

02

Keane, Neal, & Vu (2026) robust estimation & inference

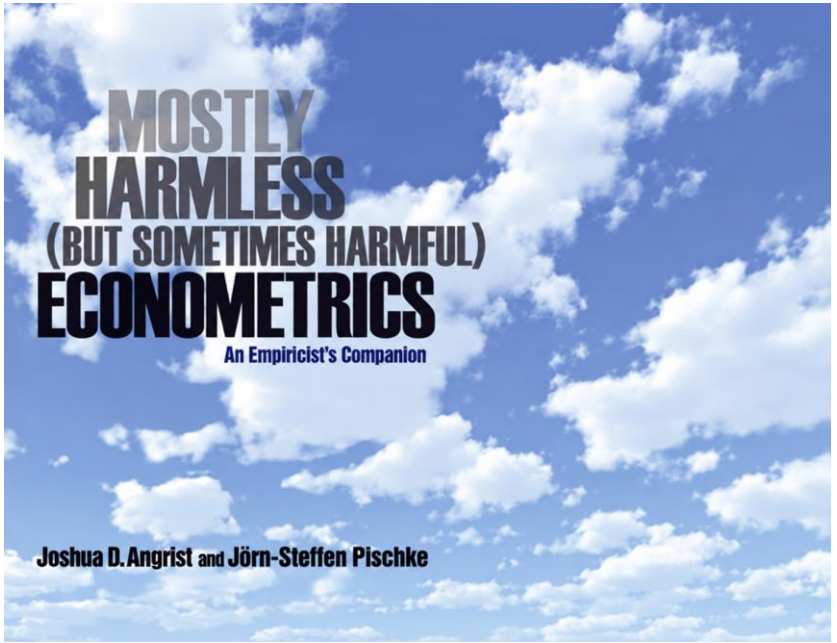
Avoid selection all together: use all available instruments with 2SLS/OGMM < FLIML for estimation and Moreira's conditional likelihood ratio test for inference (maybe a part 2 on alternative IV estimation/inference strategies?)

Let’s all be scrupulous economists!

- “Pick your single best instrument and report just-identified estimates using this one only,” (Angrist & Pischke, 2009 p. 157).

Do not do this.

Be a scrupulous economist.



Thank You

Some references

- Michael P. Keane, Timothy Neal, and Patrick Vu, "Instrument Hacking," NBER Working Paper 35633 (2026), https://doi.org/10.3386/w35633.
- Michael P. Keane, Timothy Neal. 2024. A Practical Guide to Weak Instruments. *Annual Review of Economics* 16:185-212. https://doi.org/10.1146/annurev-economics-092123-111021
- Joshua Angrist and Michal Kolesar, “One Instrument to Rule Them All: The Bias and Coverage of Just-ID IV,” *Journal of Econometrics*, 240 (2), Article 105398 (2024), https://arxiv.org/abs/2110.10556
- Alastair R. Hall, Glenn D. Rudebusch , and David W. Wilcox, ”Judging Instrument Relevance in Instrumental Variables Estimation.” (1996) Available at SSRN: https://ssrn.com/abstract=5894